1 PULL-UP RESISTORS ON DIGITAL INPUT

On the module RobDuino we can find two “on-board push button switches”. Wiring of this switches is presented in fig. 1, where can we noticed that both switches are connected to ground voltage potential.

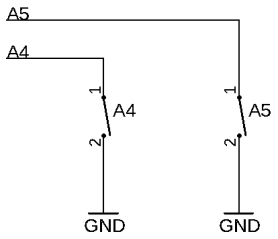

Figure 1: Wiring of on-board switches.

To properly use this on-board push-button switches we must enable the `pull-up` resistors of A4 and A5 input of microcontroller.

1.1 Tasks:

1. Configure pins `A4` and `A5` as inputs with `pull-up` resistor.
2. At the end of the `setup()` function add the **`while`**`-loop` which will delay the execution of the program until we press the `A4` key - acting as a “START BUTTON”.
3. Use the `A5` key to stop the robot and terminate the execution of the program.

Program 1: Pull Up Resistors on Digital Input.

```
#include "RobotMovingFunctions.h"
const int KEY_A4 = A4;
const int KEY_A5 = A5;

void setup()
{
  setIOpins();
  pinMode(KEY_A4, INPUT_PULLUP);
  // KEY_A5 setup here...
}

void loop()
{
  moveForward();
  //to-do: the key reading
  bool stopTheRobotKey = 0;
  if (stopTheRobotKey == 1)
  {
    stopTheRobot();
    exit(0);        //terminate the program
  }
}
```

1.2 Questions:

1. What is the programming instruction of reading the value form digital input?
2. Which values can be assigned to **`bool`** type variable?
3. Explain the programming instruction `exit(0)`.

1.3 Summary:

1.3.1 <++>

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1.4 Issues:

1.4.1 <++>

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